Sahaj Bhadja

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Summary

Software engineer with 2+ years building automated test workflows, operator-facing applications, backend services, and cloud data pipelines. Experienced in Python, Java, C++, React, SQL, AWS, Azure, simulation, CI/CD, observability, debugging, and production reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected Azure SQL and Blob Storage services for 500+ users, refining normalized schemas, indexed queries, access controls, and configuration history to reduce operator-facing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant processing for 5,000+ monthly jobs using explicit states, retries, reconciliation, and failure isolation, tripling throughput while strengthening result traceability, observability, and reliable production support.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React operator workflows and Flask REST services for secure upload, routing, status, and history tracking, translating ambiguous requirements into maintainable features supporting 150+ users and 10,000+ daily requests.
- Created concurrent processing for 1,000+ files with automated validation, pass-fail status handling, tests, and AWS RDS persistence, reducing turnaround 60% while maintaining stable application behavior during peak workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible simulation and benchmarking pipelines processing 100,000+ records, applying structured automated testing, instrumentation, debugging, and performance analysis to improve long-horizon planning accuracy 18% across controlled evaluations.
- Participated in cross-functional collaborative design reviews and validation experiments, documenting technical decisions and tuning a spatial-temporal architecture that improved closed-loop stability 15% across repeated controlled simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed object-oriented API and ETL components using reusable design patterns, AWS Glue, and S3, improving response times 30% and test-result pipeline efficiency 68% across complex production workflows.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating defect diagnosis and strengthening confidence in cross-platform production releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular React and Node.js operator workflows across scheduling, permissions, inventory-style resources, billing, and notifications, supporting 2 million+ traceable Azure SQL records through independently maintainable service boundaries.
- Automated Docker builds, validation tests, staged deployments, and production monitoring through Jenkins and Azure services, reducing release time 50% while improving software quality, traceability, and operational reliability.

Skills

Languages & Operator Tools: Python, Java, C++, TypeScript, JavaScript, React, Node.js, Flask, FastAPI, responsive interfaces, operator workflows, object-oriented programming
Test & Production Systems: test automation, simulation testing, pass-fail logic, validation, benchmarking, instrumentation, REST APIs, microservices, asynchronous workflows, concurrency, failure isolation, production support
Data & Cloud: AWS, Azure, AWS RDS, S3, Azure SQL, PostgreSQL, MySQL, MongoDB, Blob Storage, data modeling, result history, traceability, indexing, query optimization
Quality & DevOps: Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, PyTest, Mockito, unit testing, integration testing, regression testing, debugging, observability, monitoring, documentation, Agile, Scrum

Projects

CPU-Scheduling Simulator | Live | *C++, React, JavaScript, Algorithms*
- Implemented six scheduling algorithms with dynamic metrics and Gantt visualization across 1,000+ process combinations, reducing debugging time 20% through deterministic simulation, maintainable C++ logic, and systematic edge-case testing.

Chat2SQL | Code | *Java, TypeScript, SQL, Docker, Maven*
- Built a cross-platform SQL workspace supporting 10+ databases, combining Java services, schema-aware validation, Docker, and operator-friendly workflows across Windows, macOS, and Linux for reliable data inspection and management.

Big-Brain | Live | *Python, PostgreSQL, GCP, REST APIs, FAISS*
- Engineered cloud ingestion and indexed history workflows processing 50,000+ records with PostgreSQL, GCP, and REST APIs, implementing incremental updates and performance tuning to reduce retrieval latency 40%.